

TIA-942-B Facility Specification (Synthetic)

This specification dictates the operational parameters for the Tier III data center facility.

Maximum ambient temperature: 50°C (TIA-942-B §6.7.1)

UPS redundancy level: N+1 minimum (§5.2.3)

Fire suppression: FM-200 or Novec 1230 for UPS rooms >500kVA (§7.4.2)

Generator fuel consumption: ≤ 260 l/hr at rated load (§8.3.4)

Cable derating factor: 0.75 minimum for bunched cables (§9.1.2)

Cooling capacity: 500 kW minimum per unit (§10.2.1)

Floor loading capacity: 12.0 kN/m² minimum (§4.3)

Seismic bracing: Zone 4 compliance required (§4.5.1)

Chilled water supply temperature: 10°C maximum (§11.1)

Aisle containment: Cold aisle containment mandatory (§11.4)

PDU efficiency: $\geq 98\%$ at 50% load (§12.2)